

Detailed Notes: Gemini Parameter Tuning, API, SDK and Function Calling

Top-K vs Top-P Sampling

Top-K Sampling

- Limits token selection to the K most probable next tokens.
- Lower K = more deterministic output.
- Higher K = more creative output.
- Useful when you want controlled and predictable responses.

Top-P (Nucleus) Sampling

- Chooses from the smallest set of tokens whose cumulative probability exceeds P.
- Dynamic token pool size depending on context.
- Produces more natural and context-aware responses.
- Common values: Top-P = 0.8–0.95.

Comparison

Top-K uses a fixed number of candidates, while Top-P uses a probability threshold. Top-P is generally preferred for conversational AI because it adapts to the distribution of probabilities.

Combining Parameters in Gemini

Gemini generation quality depends on multiple parameters working together: • Temperature controls randomness. • Top-K restricts candidate tokens. • Top-P controls cumulative probability. • Maximum Output Tokens limits response length. • Safety settings control harmful outputs.

Recommended workflow: 1. Set a reasonable temperature. 2. Adjust Top-P and Top-K for creativity. 3. Define output length. 4. Evaluate results and iterate.

Best Practices for Parameter Tuning

1. Start with default values before making changes.
2. Tune gradually and modify one parameter at a time.
3. Test multiple prompts and scenarios.
4. Balance creativity with factual accuracy.
5. Keep records of parameter combinations and outcomes.
6. Use evaluation metrics for consistency.

Hands-on Parameter Tuning Activity

Typical tuning workflow: • Define objective (creative writing, coding, research, chatbot). • Create test prompts. • Modify temperature, Top-K, and Top-P. • Compare outputs. • Measure quality, accuracy, and consistency. • Select optimal configuration for deployment.

Real-World Applications of Parameter Tuning

Chatbots: • Improve conversational quality and user satisfaction.
Content Generation: • Control creativity and diversity of generated text.
Coding Assistants: • Increase code correctness and reduce hallucinations.

Research: • Produce balanced and evidence-based summaries.
Creative Writing: • Generate imaginative and engaging content.
Customer Support: • Maintain consistency while preserving natural responses.

Setting Up Gemini API

Step 1: Create a Google Cloud Project.
Step 2: Enable Vertex AI API or Gemini-related services.
Step 3: Install required SDKs (Python SDK).
Step 4: Configure authentication credentials.
Step 5: Connect to Gemini models and start making requests.

Important Concepts: • API Key management.
• Authentication and authorization.
• Project quotas and billing.
• Monitoring and logging.

What is Gemini Python SDK?

The Gemini Python SDK allows developers to interact with Gemini models directly from Python applications.

Features: • Simple integration.
• Prompt handling.
• Model configuration.
• Streaming responses.
• Function calling support.
• Access to multiple Gemini models.

Benefits: • Faster development.
• Cleaner code structure.
• Easier maintenance and scalability.

Function Calling in Gemini

Function Calling enables Gemini to invoke external tools and services instead of only generating text.

Examples: • Weather applications.
• Database queries.
• APIs and web services.
• Automation systems.
• AI assistants.

Advantages: • Access to real-time data.
• Reduced hallucinations.
• Improved reliability.
• Structured outputs.

How Function Calling Works

1. User submits a query.
2. Gemini identifies the user's intent.
3. Gemini decides which function/tool is required.
4. The function executes externally.

5. Structured results are returned to Gemini.
6. Gemini generates the final response using those results.

Benefits:

- Real-time information access.
- Better automation.
- Improved accuracy and decision making.
- Easier integration with enterprise systems.